

Face Recognition Emotion Detection using Artificial Intelligence

FINAL YEAR PROJECT PROPOSAL Session
2022-2026

A project submitted in partial fulfilment of the
Government College University Faisalabad
Degree Of
Bs in Computer Science



Department of Computer Science
Government Graduate College Jhang

30 April 2026

Project Registration

Project ID (for office use)			
Type of project	☐ Traditional	☒ Industrial	☐ Continuing
Nature of project	☒ Development		☐ Research
Area of specialization	Computer Vision, Artificial Intelligence		
Project Group Members			
Sr.#	Name	Roll No	Signature
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Declaration: FYP group members have cleared all prerequisite courses for FYP-I as per their degree requirements.			

Plagiarism Free Certificate

This is to certify that, I am _____ S/D/o _____, group leader of FYP under registration no _____ at the Computer Science Department, Government Graduate College, Jhang. I declare that my supervisor checks my FYP proposal and the similarity index is _____% that is less than 20%, an acceptable limit by HEC. The report is attached herewith as Appendix A.

Date: _____ Name of Group Leader: _____ Signature: _____

Name of Supervisor: _____ Co-Supervisor (if any): _____

Designation: _____ Designation: _____

Signature: _____ Signature: _____

Project Abstract

This project is about making a web app that can detect human emotions from a face photo or camera video. The system will find the emotion like happy, sad, angry, or surprised. It will use TensorFlow/ Keras ,OpenCV and Flask for the web app. This project helps in education, health, and customer service. It shows how AI can understand human emotions.

1.Introduction

In the past, people understood emotions by looking at faces, which was slow and not always correct. Now, Artificial Intelligence can detect emotions automatically using cameras. Our project, Facial Recognition Emotion Detection using Artificial Intelligence, finds emotions like happy, sad, or angry from a face. It is faster, easier, and more accurate than old manual methods.

2. Problem Statement

- Computers cannot easily understand human feelings
- People make mistakes
- Old system slow
- Need smart AI based System Recognized Facial emotions

3.Objectives

- Make a web app using Flask and Python.
- Use Artificial Intelligence to detect emotions like happy, sad, or angry.
- Show real-time emotion results from a camera or photo.
- Keep the system simple, fast, and easy to use.
- To save time and reduce mistakes.

4. Tools and Technologies

Component	Tool/Technology
Framework	Python (Flask)
Platforms	Web Application
Face Recognition	TensorFlow Lite / Keras
Image Processing	Open Cv
Database	SQLite
IDE	Pycharm / VS Code

5. Key Features

- i. **Image Module** Takes photo or video from camera.
- ii. **Face Module**– Finds the face in the image.
- iii. **Emotion Module** Detects emotion like happy, sad, or angry.
- iv. **Result Module** – Shows the emotion on the screen.
- v. **Database Module** – Saves data if needed.
- vi. **Web Module** – Simple web page made with Flask.

6. Scope of the Project

The project Facial Emotion Recognition using Artificial Intelligence can be used in many areas. It helps to detect people's emotions like happy, sad, or angry in real time. It can be useful in schools, offices, hospitals, and customer care systems. Because it is a web app, it is simple to use on any computer or laptop.

7. Methodology

- i. **Collect Data** – Take face images from datasets like FER2013 or Affect Net.
- ii. **Preprocess Data** – Crop face, resize image, and make it clear for training.
- iii. **Build Model**– Use CNN to learn face and emotion features.
- iv. **Train Model** – Give images to CNN to learn emotions (happy, sad, angry etc.)
- v. **Test Model**– Check how well it predicts emotions.

- vi. **Deployment** – Make a small app or program to show real-time results.
- vii. **Documentation** – Prepare final report.

8. Timeline

Phase	Duration	Description
Requirement Analysis	2 Week	Collect requirements
Design	2 Week	Design web pages and model structure.
Dataset Collection	1 Week	Collect and prepare emotion images.
Development	4 Weeks	Make Flask web app and connect trained model.
Testing	2 Week	Test emotion detection and fix errors.
Finalization	2 Week	Documentation & submission

9. Expected Outcome

- I. A working web app that detects human emotions.
- II. Emotions like happy, sad, and angry will be shown on the screen.
- III. The system will work fast and give correct results
- IV. It will be easy to use on any computer or laptop.

10. Conclusion

The project Facial Recognition Emotion Detection using Artificial Intelligence helps computers understand human feelings. It can detect emotions like happy, sad, or angry from a face in real time. This system is simple, fast, and useful for schools, offices, and hospitals. It shows how Artificial Intelligence can make machines smarter and more human-like.

11. Related Searches

- Facial Emotion Recognition using AI
- Artificial Intelligence for emotion detection 2025
- Human face emotion recognition system
- Face and emotion recognition using CNN

12. References

- **Deep Face Library** – AI based face and emotion (2025).
- **Emotion Detection using FER2013 Dataset** – .TensorFlow/Keras implementation(2025)
- **Facial Expression Recognition with AI (PyTorch)** – FER2013 & CK+ datasets (2024).
- **Face Classification** – Real-time emotion detection with OpenCV (2024).
- **Grad-CAM Emotion CNN** – visualization for AI systems (2025).